Henry Jochaniewicz

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EDUCATION

University of Notre Dame | Notre Dame, IN — (Expected) May 2027
Bachelor of Science | Major: Computer Science | Minors: Theology, Mathematics — GPA: 4.0

Relevant Coursework: High-Performance Distributed Systems, Operating System Principles, Compilers and Language Design, Computer Networks, Computer Architecture, Theory of Computing, (Abstract) Algebra, Topology

WORK EXPERIENCE & LEADERSHIP

Sorin College | Notre Dame, IN — August 2026 – Present
Resident Assistant

- Serve a dorm community of 150+ university men through engagement opportunities, conflict resolution, and a meaningful presence

Volexity, Inc. | Silver Spring, MD — May 2026 – August 2026
Software Engineering Intern

- Performed memory forensics with proprietary and open-source tools locally and on virtual machines and cloud instances
- Added Linux file birth time collection feature to flagship memory collection product in Go, profiling feature performance and modifying Python tests
- Debugged longstanding timestamp floating-point bug in testing infrastructure, eliminating consistent false negatives
- Researched methods and enhanced internal Go tooling to increase coverage of memory forensics profiles across Linux distributions
- Delivered comprehensive presentation to company management on project process, milestones, results, and next steps

École Centrale de Lyon | Lyon, France — May 2025 – July 2025
Undergraduate Researcher

- Extracted performance metrics from 11+ papers on AES hardware accelerators to compare to an under-development AES chip, demonstrating discovered trends by generating 15+ graphs with Python and automating the process with Google Colab
- Co-described an application of the 1FeFET-1C cell, an emerging memory technology, through a modified bit cell for the SubBytes step of encryption to improve accelerator area and performance
- Presented formal summary of contributions to 30 PhD students and advisor, with generated graphs shown at sponsor presentations

College of Engineering | Notre Dame, IN — August 2024 – Present
Undergraduate Teaching Assistant

- Provided tailored feedback and office hours for 50 students in Discrete Mathematics, 100+ students in Fundamentals of Computing, and 50+ students in Operating System Principles

Domer Rover Engineering Design Club | University of Notre Dame — September 2023 – Present
Sub-Team Lead: Radio and Communications; Team Member

- Directed sub-team of 4 members to build communications system for the University Rover Challenge with 9 other leads, spending 4–15 hours/week to architect a dual-band wireless connection at distances up to 1 km
- Designed modular, multithreaded Python serial networking protocol over serial 900MHz to transfer photos, driving controls, and status signals over sockets with techniques such as weighted round-robin scheduling and drop-recovery acknowledgments

TECHNICAL PROJECTS

Version Control System (VCS) | Personal Project — May 2026 – August 2026

- Recreated a simplified persistent VCS à la Git in Rust that supports staging, committing, branching, diffing, and checkouts

Chip-8 Emulator | Personal Project — December 2025 – January 2026

- Built a Chip-8 emulator in C and SDL3 with cycle-accurate instruction loops and plays ROMs with a real-time debugging mode

B-Minor Compiler | Course: Compilers and Language Design — September 2025 – December 2025

- Handwrote a multi-stage compiler for a C-like statically typed language which parses input into an abstract syntax tree via tools Flex and Bison, resolves variables, type-checks, and generates x86-64 assembly compatible with the System V ABI

Interpreters | Personal Project — May 2025 – August 2025

- Developed a recursive-descent interpreter in Java for a dynamic, object-oriented language to learn about programming language implementations from the textbook *Crafting Interpreters* by Robert Nystrom
- Coded a stack-based interpreter in C for the same language that single-pass compiles to bytecode and implements mark-and-sweep garbage collection, string interning, and NaN boxing to yield a 3.39x speedup

SKILLS, LANGUAGES, & HOBBIES

Technical: C, Python, Go, Rust, Java, GLSL, Bash, Assembly (x86-64, RISC-V), SDL3, Flex, Bison, Linux, Git, Vim, Virtual Machines
Languages: English (fluent), Italian (proficient), Western Armenian (basic)
Hobbies: Origami, piano, cult-classic films (e.g., Hitchcock, Studio Ghibli), reading, video games (e.g. *Outer Wilds*), traveling, writing